

Yes. I can build the complete assignment.

The project will include all the requirements from your assignment:

- Sport selection (Cricket, Football, Tennis, Basketball, etc.)
- Difficulty selection (Easy/Medium/Hard)
- AI-generated MCQs
- ChromaDB vector database (RAG)
- Web search for latest sports facts
- Quiz regeneration
- Correct answers with explanations
- Clean UI
- README and documentation

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Tech Stack

- **Frontend:** Streamlit
- **Backend:** Python + FastAPI
- **LLM:** OpenAI GPT (or Gemini if you prefer)
- **RAG:** LangChain
- **Vector DB:** ChromaDB
- **Embeddings:** OpenAI Embeddings or Sentence Transformers
- **Web Search:** Tavily Search API
- **Environment:** Python 3.11+

Project Structure

```
sports-quiz-agent/  
|  
├─ app.py                # Streamlit UI  
├─ api.py                # FastAPI backend  
├─ requirements.txt  
├─ README.md  
|  
├─ data/  
|   ├── cricket.txt  
|   ├── football.txt  
|   ├── badminton.txt  
|   └── tennis.txt  
|  
├─ rag/  
|   ├── ingest.py  
|   ├── retriever.py  
|   └── quiz_generator.py
```



```
|   └─ web_search.py
|
├─ chroma_db/
|
├─ prompts/
|   └─ quiz_prompt.txt
|
└─ utils/
    └─ helpers.py
```

Features

Dashboard

- Sport dropdown
- Difficulty dropdown
- Generate Quiz button
- Regenerate Quiz button

Output

Sport: Cricket



Difficulty: Hard

Question 1

...

A.

B.

C.

D.

Correct Answer

Explanation

AI Workflow

User



|

Choose Sport + Difficulty

|

Retrieve documents from ChromaDB

|

Search Web (latest information)

|

Merge retrieved context

|

Send context to GPT

|

Generate MCQs

|

Display quiz

RAG Pipeline

Sports Documents

|

Text Splitter

|

Embeddings

|

ChromaDB

|

Retriever

|

Recent Web Search

|

LLM

|

Quiz Generation



Required APIs

OpenAI

OPENAI_API_KEY=



or

Gemini

GOOGLE_API_KEY=



Tavily

TAVILY_API_KEY=



Expected Output

Sport : Badminton



Difficulty : Medium

Question 1

Which country won the Thomas Cup in 2022?

A. Indonesia

B. India

C. China

D. Denmark

Correct Answer

B. India

Explanation

India defeated Indonesia in the final to win its first Thomas Cup.

Deliverables

- Working Streamlit dashboard
- FastAPI backend
- ChromaDB integration

-  RAG implementation
-  Web search integration
-  Quiz regeneration
-  README
-  Well-documented code
-  GitHub-ready project structure

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I can generate the entire project (around 700–1000 lines of code) with all files ready to run. Because it exceeds the maximum response length, I'll provide it in multiple parts:

1. Project setup and `requirements.txt`
2. ChromaDB ingestion and retrieval
3. Web search integration
4. Quiz generation agent
5. FastAPI backend
6. Streamlit frontend
7. README with setup and run instructions

This will result in a complete, submission-ready project.